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Measles resurgence in Bangladesh, 2026: a warning of fragile immunization gains

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Measles outbreak in Bangladesh in 2026

Bangladesh is facing a major measles outbreak, signalling a reversal of previous progress toward elimination. Between 15 March and 29 April 2026, 35,980 suspected cases, 4,944 laboratory-confirmed cases, 227 suspected measles-related deaths, and 47 confirmed measles-related deaths were reported (**Fig 1**), resulting in a case-fatality rate of 0.63% (227/35,980) among the suspected cases and 0.95% (47/4,944) among the confirmed cases¹. The outbreak affected 61 of 64 districts across all eight divisions, with the highest burden in the central Dhaka division (13,685 cases), northern division Rajshahi (5,832 cases), eastern division Chattogram (4,065 cases), and southern division Khulna (2,337 cases). Children aged under five years accounted for 79% of reported cases¹. The daily number of suspected measles cases remains consistently high, with weekly cumulative cases reaching 8,198 in week 17 (starting at 19th April 2026), indicating sustained transmission and ongoing expansion of the outbreak. (**Fig 2**). The outbreak has mostly affected densely populated urban areas in most districts². Hospitals are under significant strain, with overcrowded wards, patients treated in corridors, and reports of early discharge and readmission due to limited capacity and resources³.

Bangladesh has made substantial progress and gained international recognition for the success of its Expanded Programme on Immunization (EPI)⁴. Since its introduction in 1979, the programme has achieved remarkable gains, increasing the proportion of fully immunized children from just 2% to 81.6%⁴. It now saves an estimated 94,000 lives and prevents around 5 million childhood illnesses each year, generating an approximate return of \$25 for every US\$1 invested⁴.

Measles is a highly contagious infection transmitted through respiratory droplets and airborne spread, with the virus remaining infectious in the air up to two hours. It typically presents with fever, cough, coryza, and conjunctivitis, followed by a characteristic maculopapular rash. Severe complications, including pneumonia and encephalitis, are more common in young children and unvaccinated individuals². Measles is one of the most infectious human diseases. For Measles ($R_0=12-18$), the herd immunity threshold (HIT), estimated as $(1-1/R_0)$, is approximately ~92-95%⁵. In real-world settings, the required vaccination coverage must be even higher when accounting for imperfect vaccine effectiveness, clustering of unvaccinated children, and unequal access to immunization services. This explains why even modest reductions in coverage can rapidly lead to large-scale outbreaks.

So, what went wrong in Bangladesh:

In Bangladesh, routine measles vaccination is provided to children from nine months of age, supplemented by nationwide measles vaccination campaigns conducted approximately every four years³. However, no such campaigns have been implemented since 2020, initially due to the COVID-19 pandemic and subsequently because of the political situation. A planned follow-up campaign in 2024 was not implemented due to political unrest³. Shortages of health workers were also reported, with 45% of positions vacant in more than half of districts in 2025, while several senior roles in the health sector remained unfilled for months during transitions across three successive governments over the past two years, alongside restructuring of the EPI⁶. Measles vaccine coverage has dropped from about 90% in 2020 to roughly 57% in 2025³. This prolonged interruption has likely contributed to the accumulation of susceptible children, increasing the risk of large-scale outbreaks.

In April 2025, Gavi, the Vaccine Alliance, UNICEF, and the World Health Organization (WHO) reported that around 400,000 children in Bangladesh remain unimmunized, including about 70,000 (1.5%) who have received no vaccines. Coverage gaps are more pronounced in urban areas, where only 79% of children are fully vaccinated, with 2.4% zero-dose and 9.8% under-immunized, compared with 85% coverage in rural areas⁴.

Vaccine hesitancy is not a major driver of under-immunization in Bangladesh, where acceptance of routine childhood vaccination remains generally high. Instead, recent declines in coverage appear to be driven largely by supply-side and systemic constraints, including vaccine shortages, COVID-19-related disruptions to routine immunization services, interruptions to service delivery, and gaps in outreach⁷. These vulnerabilities have been further exacerbated by disruptions to vaccination activities during the 2024 student-led political unrest, compounding existing coverage gaps³.

The recent vaccination shortages have not been limited to measles-containing vaccines but have also affected other childhood vaccines delivered through the EPI, including the pentavalent vaccine Diphtheria with an R_0 of 2.5-7.2 and HIT of 60-86%; Pertussis, $R_0=(5-17)$, HIT=(80-94%); and *Haemophilus influenzae* type b infection, $R_0=(1.3-3.3)$, HIT=(23-69%), as well as Rubella $R_0=(3.0-$

7.0), HIT =(67-86%), which is co-administered with measles as MR1 and MR2 doses⁵. Although these infections generally have lower transmissibility than measles and therefore require comparatively lower herd immunity thresholds, this should not be interpreted as a low outbreak risk.

Immunity gaps can accumulate silently, particularly in densely populated and underserved settings, allowing transmission to intensify once a critical threshold is crossed. Of particular concern is pertussis, where higher estimates of transmissibility would raise the herd immunity threshold to approximately 94%, approaching that of measles⁵. Under such conditions, the same structural weaknesses that have enabled the current measles outbreak could facilitate the resurgence of other vaccine-preventable infections. The ongoing outbreak should therefore be viewed as an early warning signal rather than an isolated event. Without rapid restoration and equitable distribution of vaccination coverage, the risk extends beyond measles to a broader resurgence of childhood infectious diseases in the near future.

In response to the outbreak, Bangladesh has implemented emergency public health measures centred on a nationwide vaccination campaign. The first phase began on 5 April 2026 in 30 of the most affected upazilas (sub-districts), followed by expansion to major cities including Dhaka and Chattogram on 12 April, with nationwide rollout from 20 April³. Authorities lowered the minimum age for measles vaccination to six months, below the routine schedule of nine months, consistent with WHO recommendations for outbreak settings, to rapidly protect vulnerable infants³. The campaign aims to vaccinate up to 1.2 million children aged 6 months to 5 years, with the target age extended to 10 years in high-risk districts. To support this effort, Bangladesh secured 21.9 million doses of measles-rubella vaccine from Gavi, the Vaccine Alliance, alongside additional procurement funded by the Asian Development Bank and implemented through UNICEF³.

The current outbreak highlights the urgent need to restore and sustain high immunization coverage across Bangladesh, with a particular focus on reaching zero-dose and under-immunized children in densely populated urban settings, where the burden of disease is highest. Strengthening routine immunization systems should be prioritised alongside periodic supplementary measles and rubella campaigns to close existing immunity gaps and prevent further accumulation of susceptible cohorts. Given that recent disruptions have affected multiple components of the EPI, there is a credible risk of resurgence of other vaccine-preventable diseases, particularly pertussis and diphtheria, if gaps are not urgently addressed. The measles outbreak should therefore be treated as a sentinel event, underscoring the need for sustained investment in resilient, equitable, and integrated public health systems to safeguard past gains and prevent wider infectious disease resurgence. The COVID-19 pandemic caused widespread disruption to routine immunisation services across many low-and middle-income countries⁷, leading to the accumulation of susceptible populations and increasing the risk of outbreaks. The current measles resurgence in Bangladesh is likely to represent only one of a series of outbreaks

that may emerge in similar settings in the coming years. This underscores the need for coordinated action by WHO, UNICEF, and Gavi, the Vaccine Alliance, to prioritise restoration of routine immunization, expand catch-up vaccination campaigns, and strengthen surveillance systems in high-risk settings

Authors Contribution Statement

Conceptualization: NH, Data curation: NH, writing original draft: NH, Writing, review, and editing: RK, ZH

Conflicts of interest

The authors declare no conflict of interest. Zakiul Hassan is an Assistant Editor for *International Journal of Infectious Diseases*. The article will be published as an editorial and, as such, will not undergo a formal review process.

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Data availability

The 2026 measles case and death data for Bangladesh reported in this article are publicly available from the Ministry of Health and Family Welfare and can be accessed in Bengali at: <https://dghs.gov.bd/pages/press-releases/69f1c1a887220b917a0e5013>. Annual measles case data were obtained from the WHO measles database, available at: <https://immunizationdata.who.int/global/wiise-detail-page/measles-reported-cases-and-incidence>.

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Figure legend:

Figure 1: Reported annual measles cases in Bangladesh from 2000 to 2025 based on the WHO database (<https://immunizationdata.who.int/global/wiise-detail-page/measles-reported-cases-and-incidence>). The 2026 data are derived from the Ministry of Health daily press briefings and cover the period from 15 March to 25 April 2026.

Figure 2: The daily number of suspected measles cases in Bangladesh in 2026 shows a rapidly increasing trend. Publicly available data are available from 4 April 2026 onwards. Weekly cumulative cases rose sharply, with 1,734 cases in week 14 (starting 29 March 2026), 7,992 cases in week 15 (starting 5 April 2026), 8,024 cases in week 16 (starting 12 April 2026), and 8,198 cases in week 17 (starting 19 April 2026). In week 18 (starting 26 April 2026), 6,524 cases were reported within the first five days alone.

Figure 1

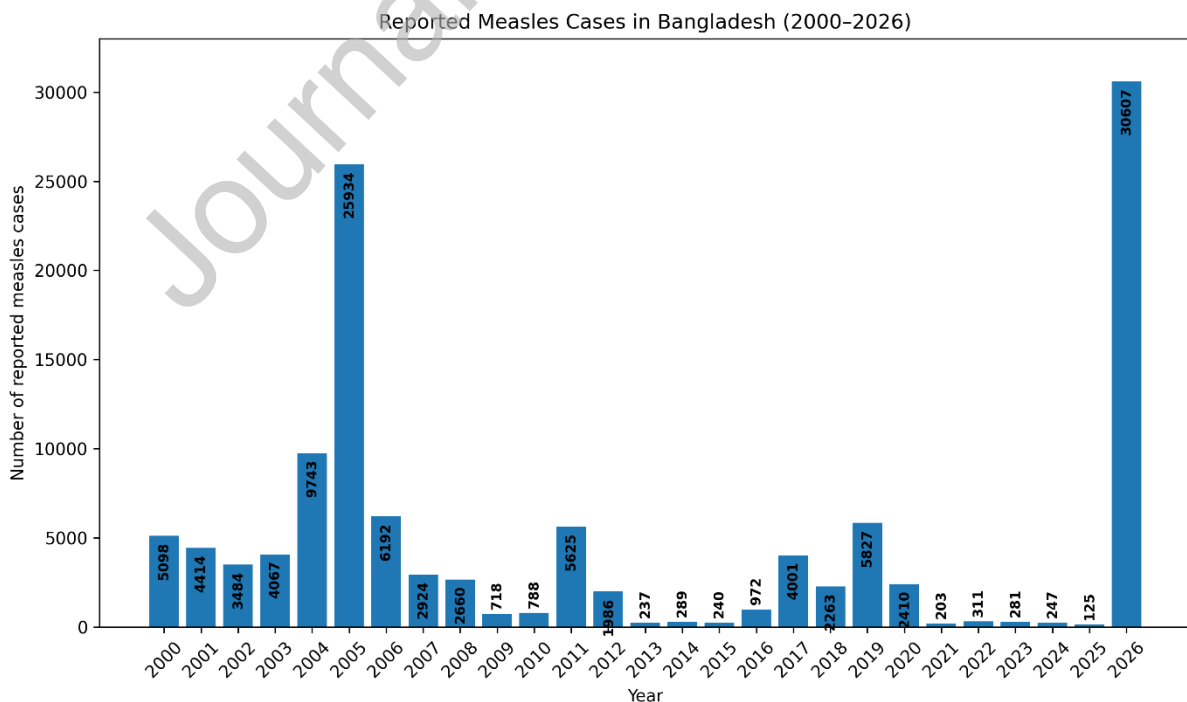


Figure 2